

CLINICAL RESEARCH

Risk Factors of Periodontitis: Evidence From Two-Sample Mendelian Randomization and Meta-Analysis

Lingjun Yan¹ | Zilin Liu² | Bingqin Xie² | Yu Huang² | Yuxuan Wu² | Baochang He²  | Yanfen Li³ | Lan Luo⁴ | Fuhua Yan³  | Fa Chen² 

¹Department of Neurosurgery, Neurosurgery Research Institute, The First Affiliated Hospital, Fujian Medical University, Fuzhou, China | ²Department of Epidemiology and Health Statistics, Fujian Provincial Key Laboratory of Environment Factors and Cancer, School of Public Health, Fujian Medical University, Fuzhou, China | ³Department of Periodontology, Nanjing Stomatological Hospital, Affiliated Hospital of Medical School, Nanjing University, Nanjing, China | ⁴Fujian Key Laboratory of Oral Diseases, Department of Periodontology, School and Hospital of Stomatology, Fujian Medical University, Fuzhou, China

Correspondence: Fuhua Yan (yanfh@nju.edu.cn) | Fa Chen (chenfa@fjmu.edu.cn)

Received: 10 February 2025 | **Revised:** 19 May 2025 | **Accepted:** 22 May 2025

Funding: This work was supported by the National Natural Science Foundation of China (82301071).

Keywords: causal inference | mendelian randomization | obesity | periodontitis | smoking

ABSTRACT

Aim: Mendelian randomization is a more appropriate tool for causal inference, as the main suspected risk factors for periodontitis are difficult to test by randomized controlled trials due to ethical or feasibility issues. This study aimed to evaluate potential causal relationships between 50 known and suspected factors and periodontitis risk by a two-sample Mendelian randomization study and meta-analysis.

Methods: By utilizing the databases of the Gene-Lifestyle Interactions at Dental Endpoints (GLIDE) consortium and the Finnish Genetics (FinnGen) consortium, 25 obesity-related indicators (BMI, birth weight, weight, height, waist-hip ratio, waist circumference, hip circumference, 18 body fat percentage or fat-free mass factors), eight hormone-related indicators (estradiol levels, total testosterone levels, sex hormone-binding globulin, age at menarche, age at menopause, three bone mineral density factors), five lifestyle factors (smoking, alcohol drinking, sleep duration, morning/evening chronotype, years of schooling), three dietary factors (coffee, tea, fruit), six blood biomarkers (fasting glucose, high-density lipoprotein cholesterol [HDL cholesterol], low-density lipoprotein cholesterol [LDL cholesterol], total cholesterol, serum 25-hydroxyvitamin D levels, hemoglobin A1c [HbA1c]), and three diseases (hypertension, type 2 diabetes, COVID-19). The odds ratios (ORs) and 95% confidence intervals (CIs) associated with the risk of periodontitis were estimated for each trait using the inverse variance weighting (IVW) method. A meta-analysis was conducted to analyze the causal associations from these databases.

Results: Among the 50 potential risk factors, the IVW analyses revealed significant associations with the risk of periodontitis for 22 and two traits (FDR-corrected $p < 0.05$) in the GLIDE database as well as the FinnGen database, respectively. The meta-analyses revealed that 23 traits maintained statistically significant associations with periodontitis risk. Noteworthy associations included 20 obesity-related indicators with ORs ranging from 1.11 to 1.25, smoking (OR = 1.74), and hemoglobin A1c (OR = 1.07), which were associated with an increased risk of periodontitis. Conversely, increased years of education (OR = 0.81) were identified as potential mitigators of periodontitis risk. The sensitivity analyses utilizing five additional methods further bolstered the robustness of these findings.

Y.-J. Yan and Z.-L. Liu contributed equally to this study.

Conclusions: This comprehensive study provides evidence for the potential causal association of several modifiable risk factors with periodontitis, highlighting the importance of addressing these factors in preventive strategies for periodontal health.

Summary

- Background
 - Numerous studies have demonstrated that periodontitis is a prevalent oral disease influenced by a multitude of factors. However, the causal association between these known and suspected factors and periodontitis remains uncertain. To date, few studies have comprehensively explored the association of multiple factors with periodontitis risk using the Mendelian randomization (MR) analysis and meta-analysis.
- Added value of this study
 - Out of 50 known and suspected risk factors, 23 traits are associated with the risk of periodontitis, of which 20 obesity-related indicators, smoking, and hemoglobin A1c show a positive association, while the level of education shows a negative association.
- Clinical implications
 - This study serves as a call for action to address these modifiable factors, highlighting the need to implement multifaceted strategies for the prevention of periodontitis.

1 | Introduction

Periodontitis is a prevalent chronic inflammatory disease characterized by the destruction of the tooth-supporting tissues, including the gingiva, periodontal ligament, and alveolar bone. It represents a significant public health concern globally, with an estimated adult prevalence of nearly 1.1 billion in 2021 [1]. Periodontitis not only affects oral health but also has implications for systemic health. There is growing evidence linking periodontitis to various systemic conditions, including cardiovascular diseases, diabetes, metabolic syndrome, and malignant tumors [2, 3]. Therefore, investigating the causal relationships between various risk factors and periodontitis is essential for elucidating the underlying mechanisms of disease pathogenesis and developing targeted interventions.

Numerous epidemiological studies have suggested potential associations of lifestyle-related factors, including smoking and diet, obesity, hormone levels, environmental exposure, with the risk of periodontitis [4–10]. Nonetheless, the causal relationships within many of these associations remain to be definitively established, as the bulk of the evidence stems from observational studies. The results of observational studies may sometimes be inevitably affected by issues such as selection bias, recall bias, confounding variables, reverse causation bias, and measurement error [11, 12]. While randomized controlled trials represent the gold standard for establishing causality, many major suspected risk factors for periodontitis cannot be tested through randomized controlled trials due to

ethical considerations (e.g., deliberately exposing participants to smoking or inducing obesity) or feasibility challenges (e.g., manipulating diabetes status or educational level). This critical gap in causal evidence necessitates alternative methodological approaches. Mendelian randomization (MR) is a method that offers a compelling methodological refinement over traditional observational studies by leveraging the random allocation of genetic variants at conception as instrumental variables. This approach circumvents key limitations inherent to observational studies, such as confounding and reverse causation biases [13]. Several recent studies have systematically explored causal associations with periodontitis, including investigations of hundreds of dietary exposures and thousands of proteins [14, 15]. However, a comprehensive investigation exploring the causal association of diverse known and suspected risk factors across multiple categories (including lifestyle, dietary, metabolic, clinical factors, etc.) remains limited. Therefore, this MR study aimed to conduct a comprehensive analysis of the association between 50 potential risk factors and the risk of periodontitis. By leveraging genetic data from large-scale genome-wide association studies, this study seeks to identify modifiable risk factors that may contribute to the risk of periodontitis, which has the potential to deepen our understanding of periodontitis etiology and guide prevention strategies for periodontitis.

2 | Methods

2.1 | Study Design

As depicted in Figure 1, we conducted a two-sample MR study and meta-analyses to investigate the association between 50 known and suspected risk factors and periodontitis. The MR analysis in this study is based on three key assumptions: (1) the genetic variants used as instrumental variables are strongly associated with the risk factors of interest, (2) the genetic variants are not associated with confounders that could bias the results, and (3) the genetic variants only affect the outcome through the risk factors being studied. This study was exempt from ethical approval from Fujian Medical University as it used publicly available data from publicly accessible databases. All data used in this study were analyzed and reported in accordance with the STROBE reporting guideline for MR studies [16]. Furthermore, our study follows the same principle framework of MR in oral health research [17].

2.2 | Instrumental Variable for 50 Potential Risk Factors

Through a meticulous review of the existing literature, we curated a comprehensive list of 50 potential risk factors for periodontitis, spanning six categories: 25 obesity-related indicators (BMI, birth weight, weight, height, waist-hip ratio, body fat percentage, whole

body fat mass, whole body fat-free mass, trunk fat mass, trunk fat percentage, trunk fat-free mass, two arm fat mass, two arm fat-free mass, two arm fat percentage, waist circumference, hip circumference, two leg fat mass, 2 leg fat-free mass, 2 leg fat percentage), eight hormone-related indicators (estradiol levels, total testosterone levels, sex hormone-binding globulin, age at menarche, age at menopause, total body bone mineral density, heel bone mineral density, ultradistal forearm bone mineral density), five lifestyle factors (ever smoked, alcohol intake frequency, sleep duration, morning/evening chronotype, years of schooling), three dietary factors (coffee intake, tea intake, fruit intake), six blood biomarkers (fasting glucose, HDL cholesterol, LDL cholesterol, total cholesterol, serum 25-hydroxyvitamin D levels, hemoglobin A1c [HbA1c]), and three diseases (hypertension, type 2 diabetes, COVID-19). Genetic variants associated with these traits were retrieved from the IEU open GWAS database (<https://gwas.mrcieu.ac.uk>) [18, 19].

To strengthen the relevance of risk factors, we have taken the following steps. First, the genome-wide significant threshold of $p < 5 \times 10^{-8}$ was established for the selection of highly associated genetic variants. To ensure the adequacy of genetic variants for our analysis, we finally employed a lenient criterion for inclusion, with the threshold for genetic variants set at 5×10^{-6} . Second, the exclusion criteria of instrumental variables that were in a state of linkage disequilibrium (LD) were set to $kb = 5000$ and $r^2 < 0.01$. A detailed information of the instrumental variables for each trait is available in Table S1. Additionally, outliers in IVs were addressed using the Mendelian Randomization Pleiotropy RESidual Sum and Outlier (MR-PRESSO) approach. The strength of the selected SNPs is evaluated by calculating the F-statistic, and the formula is as follows: $F = R^2 \times (N - 2) / (1 - R^2)$, where R^2 represents the phenotypic variance explained by each genetic variant in relation to the cytokine exposure and N is the sample size. As shown in Table S1, F-statistics greater than or equal to 10 are considered for inclusion in SNPs.

2.3 | Data Sources for Periodontitis

The summary-level statistic data for periodontitis were sourced from two databases, as delineated in Table S2. These datasets were derived from the gene-lifestyle interactions in dental endpoints (GLIDE) consortium [20], as well as the FinnGen consortium [21]. In the GLIDE consortium, periodontitis was diagnosed using definitions applied by each participating cohort, which were based on Centers for Disease Control and Prevention/American Academy of Periodontology (CDC/AAP) case definition or Community Periodontal Index based on probing depths and the number of deep periodontal pockets or self-reported of periodontitis [22]. The Community Periodontal Index based on probing depth and the number of deep periodontal pockets was defined as two or more tooth surfaces with a probing depth of ≥ 5 mm, or at least four tooth surfaces with a probing depth of ≥ 4 mm; or a probing depth of ≥ 5.5 mm in two or more sextants. Periodontitis diagnosis in FinnGen was based on ICD8 code 5234 (chronic periodontitis), ICD9 code 5234 (chronic periodontitis) and ICD10 K05.30 (chronic periodontitis, not specified) and K05.31

(chronic periodontitis, complexity) in the Finnish version of the International Classification of Diseases (ICD) [21].

The GWAS data pertaining to periodontitis from the GLIDE consortium encompassed 17 353 cases and 28 210 controls. Notably, the GLIDE consortium represents an epidemiological cohort collection characterized by comprehensive information on clinical endpoints of dental disease. Additionally, the GWAS data concerning periodontitis from the FinnGen consortium featured 3064 cases and 195 395 controls. The overarching aim of the FinnGen consortium is to elucidate the intricate interplay between genes and health by amassing genetic and health-related information from the Finnish population.

2.4 | Statistical Analyses

In order to ensure consistency in the impact of SNPs on both exposure and outcome alleles, the datasets related to traits of interest and periodontitis were meticulously harmonized. The primary MR analysis utilized the inverse variance weighted (IVW) method to assess the causal effects in terms of odds ratios (OR) and 95% confidence intervals (CI) for the association between 50 potential risk factors and periodontitis. The IVW method aggregates Wald estimates from individual SNPs to derive an overall effect estimate, with the robustness of the findings reliant on the absence of horizontal pleiotropy for each SNP. Meta-analyses were conducted to combine the results of MR analyses based on two periodontitis databases, with the utilization of the “meta” package (version 6.5–0). Consistent with the guidance from the Cochrane Handbook, the choice of whether to use a fixed-effect model or a random-effects model was not based solely on heterogeneity statistics; rather, we employed a fixed-effect model when there was an assumption of a common genetic effect across the two databases, and a random-effects model when we suspected that the genetic effects might vary. Although statistical heterogeneity (as measured by Cochran's Q test and I^2) was also assessed, it did not solely determine model choice. False discovery rate (FDR) correction was applied to the p values based on the Benjamini–Hochberg method using the `p.adjust` function of the “stats” package (version 4.3.1). Bubble plots were generated to visually represent the effects of the potential risk factors for periodontitis the “ggplot2” package (version 3.4.3).

Furthermore, a series of sensitivity analyses were carried out using four analytical approaches, including MR Egger regression (MR-Egger), weighted-median, simple mode, and weighted mode. To avoid genetic variants in similar metrics were associated with only one specific factor, we chose similar factors (arm fat mass, arm fat percentage, leg fat mass, leg fat percentage, leg fat-free mass) and removed the same SNPs from both exposures (right and left) of each metric before analyzing them with periodontitis results by IVW. In addition, we utilized PhenoScanner (website: www.phenoscanter.medschl.cam.ac.uk) to examine if these SNPs had any connections with the potential influencing factors such as smoking, alcohol consumption, level of physical activity, body mass index (BMI), and diabetes. SNPs related to any of these potential confounders were removed [23]. The association between the impact of genetic variants on exposure and outcome was illustrated using scatter plots. Heterogeneity

among genetic variants was assessed using Cochran's Q test. Horizontal pleiotropy was evaluated through the MR-PRESSO global test and MR-Egger intercept, with a *P*-value greater than 0.05 indicating the absence of horizontal pleiotropy. The statistical analysis was carried out using R (version 4.3.1), with the utilization of the "TwoSample" package (version 0.5.7) and "MR-PRESSO" package (version 1.0) during the analysis. All statistical tests were two-tailed and considered statistically significant at a significance level of 0.05.

3 | Results

3.1 | Influence of Potential Risk Factors on Periodontitis Using GLIDE Database

In our investigation of 50 potential risk factors using IVW analysis, we identified significant associations between 22 factors and the risk of periodontitis, as shown in the forest plot in Figure S1 and the red bubble in Figure S3A. Notably, within these factors,

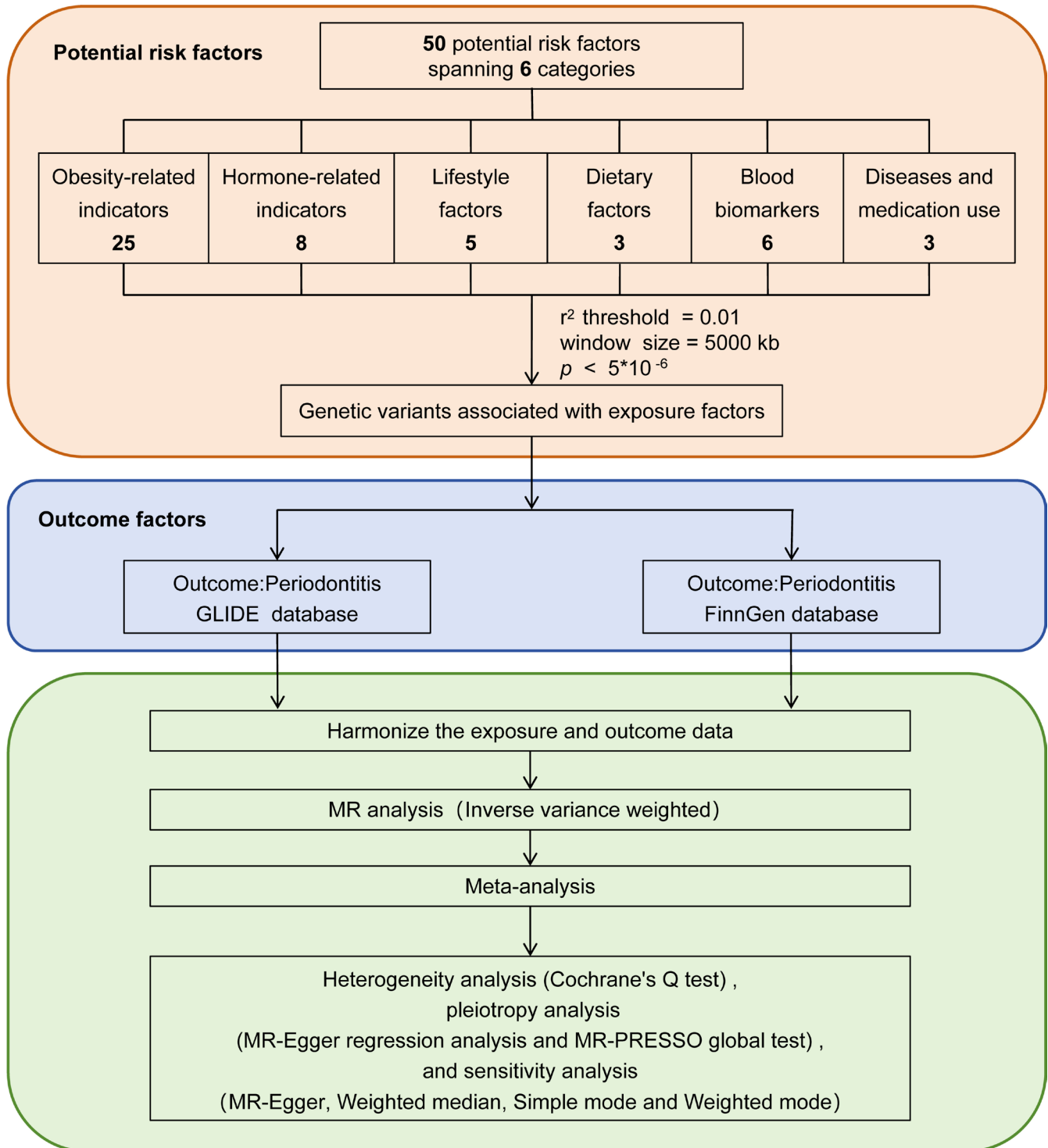


FIGURE 1 | Flowchart of study design.

obesity-related indicators accounted for 17, with increased BMI, weight, hip circumference, body fat percentage, whole body fat mass, two trunk fat indicators, four arm fat indicators and six leg fat indicators all demonstrating a significant positive correlation with the onset of periodontitis (FDR-corrected $p < 0.05$), with ORs ranging from 1.11 to 1.25.

Additionally, in our analysis of lifestyle and dietary factors, smoking was found to significantly increase the risk of periodontitis (OR = 1.97, 95% CI = 1.19–3.27). A similar positive association was observed between tea consumption and periodontitis (OR = 1.54, 95% CI = 1.15–2.07). Moreover, a longer duration of education was found to lower the risk of periodontitis (OR = 0.85, 95% CI = 0.75–0.97). The association of all variables with the risk of periodontitis can be visualized in Figure S1.

3.2 | Influence of Potential Risk Factors on Periodontitis Using FinnGen Database

In the analysis utilizing the FinnGen database with the IVW method, as illustrated in Figures S2 and S3B, a significant association was observed between 15 potential risk factors and the risk of periodontitis. Among these, 11 obesity-related indicators exhibited statistically significant associations with periodontitis, including BMI, waist circumference, trunk fat mass, whole body fat mass, four arm fat indicators, and three leg fat indicators. After correction for FDR, BMI (OR = 1.25, 95% CI = 1.12–1.39) and years of schooling (OR = 0.73, 95% CI = 0.61–0.88) were significantly correlated with the onset of periodontitis, and these results are consistent with those of the GLIDE database.

Furthermore, a higher level of education exhibits an inverse relationship with the risk of periodontitis (OR = 0.73, 95% CI = 0.61–0.88, FDR corrected $p = 0.023$). The association of all variables with the risk of periodontitis can be visualized in Figure S2.

3.3 | Meta-Analyses of GLIDE and FinnGen Databases

Following the amalgamation of data from the GLIDE and FinnGen databases, meta-analyses revealed that 23 factors remained a statistically significant association with the risk of periodontitis. Notably, among the factors linked to an increased risk of periodontitis were 20 obesity-related indicators (including BMI, weight, waist circumference, hip circumference, 3 whole body fat indicators, 2 trunk fat indicators, 5 arm fat indicators, and 6 leg fat indicators), smoking, and HbA1c. Conversely, factors such as years of education were identified as potentially reducing the risk of periodontitis (Figure 2). Among these factors, smoking emerged as the most significant contributor to periodontitis risk, followed by obesity-related indicators (Figure 3). Additionally, the relationships between other factors and the risk of periodontitis did not exhibit statistical significance.

3.4 | Heterogeneity and Sensitivity Analyses

The use of Cochrane's Q test for heterogeneity analysis did not reveal significant evidence of heterogeneity in the impact of the

mentioned exposure factors on the risk of periodontitis ($p > 0.05$, Table S3). Furthermore, through MR-Egger regression analysis to assess the intercept terms of SNPs for investigating the presence of directional pleiotropy, as shown in Table S4, the intercept terms showed no significant bias from zero, indicating the absence of directional pleiotropy for the exposure factors (all intercept terms, $p > 0.05$). Additionally, the MR-PRESSO global test did not detect any substantive level of pleiotropy for these factors (all $p > 0.05$). Additionally, employing four sensitivity analysis methods, including MR-Egger regression, Weighted-median, Simple mode, and Weighted mode, the results obtained were largely consistent with the association direction and magnitude presented by the IVW method (Tables S5 and S6).

Further analyses were performed due to the high correlation with certain risk factors in the obesity-related indicators. We screened for comparable indicators, that is, arm fat mass, arm fat percentage, leg fat mass, leg fat percentage, and leg fat-free mass. The same SNPs were excluded and then analyzed using the IVW method. Among the 16 exposures screened, only right leg fat mass (OR = 1.26, 95% CI = 1.01–1.56), left leg fat mass (OR = 1.25, 95% CI = 1.00–1.55) and right leg fat percentage (OR = 1.31, 95% CI = 1.03–1.67) showed the presence of a potential risk relationship with periodontitis. Information on the results can be found in Table S7.

4 | Discussion

Our study, utilizing a two-sample MR approach and meta-analyses of data from the GLIDE and FinnGen databases, sheds light on the causal relationship between various potential risk factors and periodontitis. The findings reveal a significant statistical association between 23 factors and the risk of periodontitis. Notably, obesity-related factors and smoking were identified as contributors to an increased susceptibility to periodontitis. Conversely, higher educational attainment was found to be associated with a reduced risk of periodontal disease. This comprehensive analysis underscores the multifaceted nature of periodontitis etiology and emphasizes the importance of addressing modifiable risk factors in the prevention and management of this prevalent oral health condition.

The complex interplay between obesity and periodontitis has been a subject of extensive investigation, with numerous epidemiological studies suggesting a positive association [24, 25]. However, the causal nature of this relationship has remained elusive, partly due to the inherent limitations of observational studies, such as confounding and reverse causation. In this two-sample MR study, we leveraged genetic instruments to dissect the potential causal links between 20 obesity-related traits—including BMI, weight, waist circumference, hip circumference, three overall body fat measures, two trunk fat measures, five arm fat measures, and six leg fat measures—and the risk of periodontitis. Notably, all 20 obesity-related traits displayed a positive association with periodontitis, underscoring the systemic implications of adiposity on oral health. The mechanistic underpinnings of how obesity might increase the risk of periodontitis are multifaceted. Systemic inflammation is a hallmark of obesity, characterized by elevated levels of pro-inflammatory cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-6

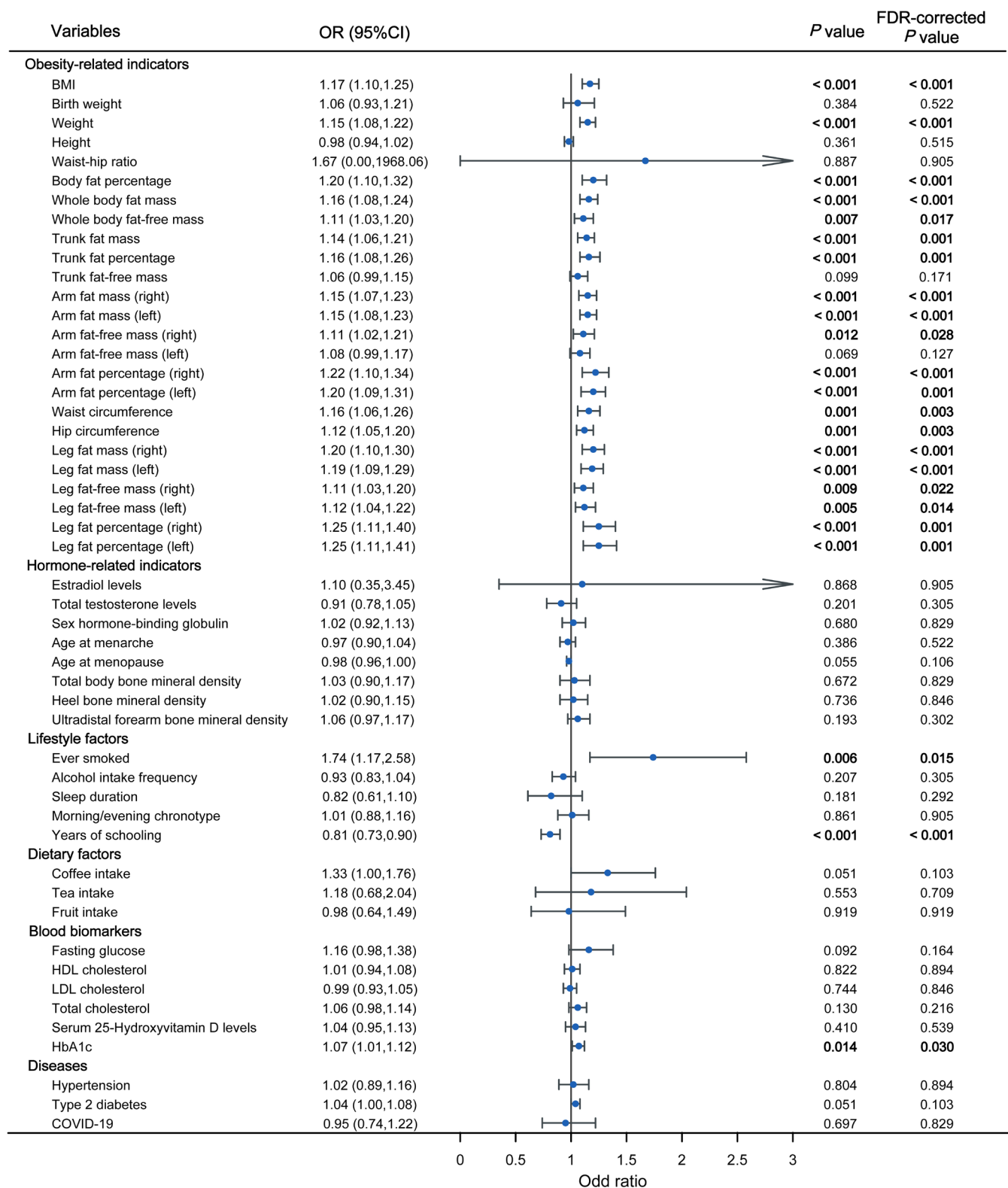


FIGURE 2 | Forest plot of risk factors significantly associated with the risk of periodontitis based on the meta-analysis of GLIDE and FinnGen databases.

(IL-6) [26], which can exacerbate periodontal tissue destruction. Moreover, adipose tissue is metabolically active and produces a variety of adipokines, including leptin and adiponectin [27], which may influence the periodontal microenvironment and enhance the risk for compromised periodontal healing [28]. Furthermore, obesity can lead to altered oral microbiota

composition, potentially favoring a dysbiotic community that fosters periodontal pathogenesis [29, 30]. Additionally, physiological changes associated with obesity, such as impaired lipid metabolism and insulin resistance, may also impact periodontal inflammation and alveolar bone loss [31, 32], given the close interaction between periodontal health and systemic metabolism.

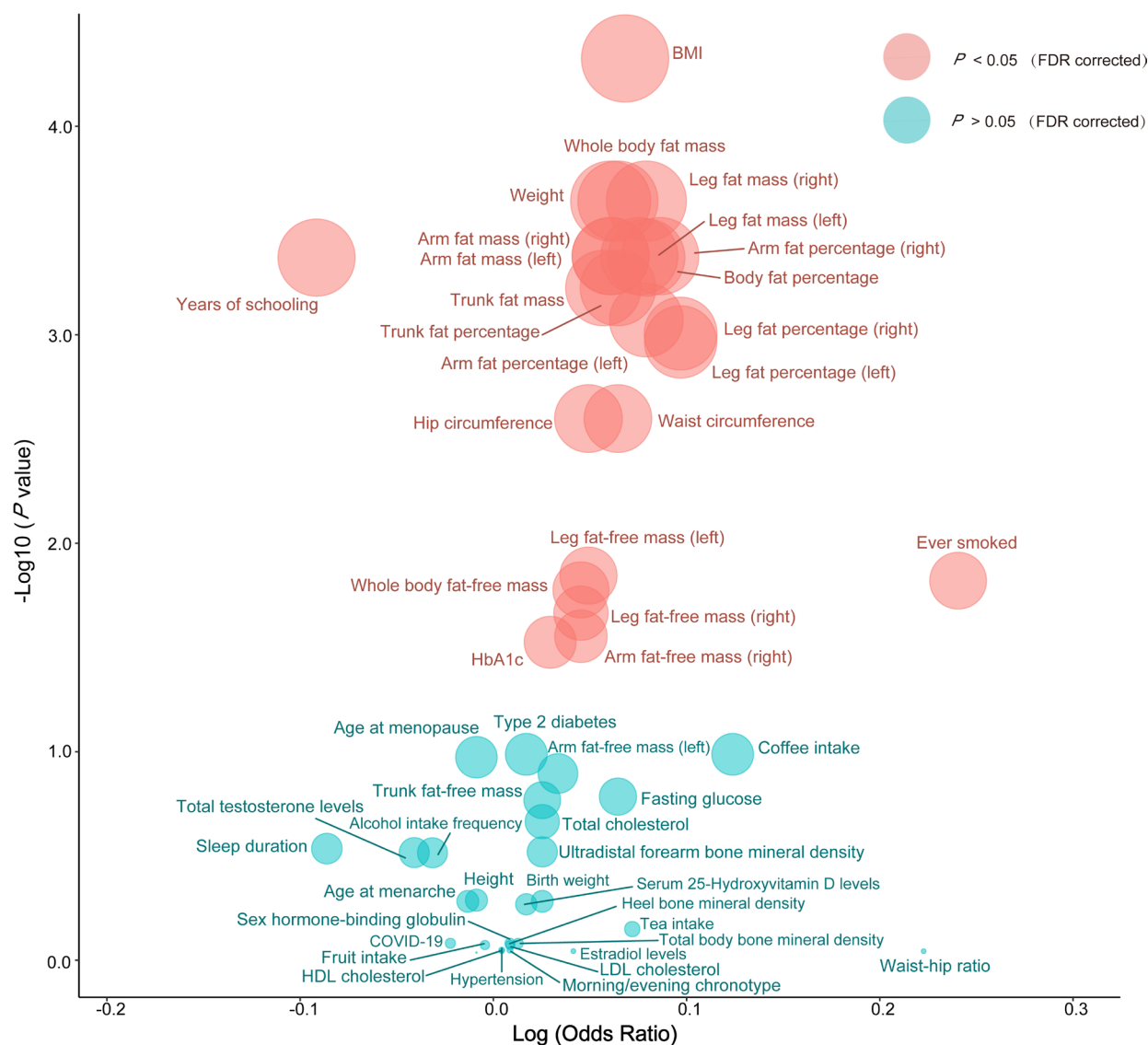


FIGURE 3 | Bubble chart summarizing the association between 50 known and suspected factors and periodontitis based on the meta-analysis of GLIDE and FinnGen databases. The x-axis illustrates the log(odds ratio) for the association between different traits and periodontitis risk, while the y-axis represents the $-\log_{10}(p \text{ value})$ for each association. Traits with a false discovery rate (FDR)-corrected p value below 0.05 are denoted by red bubbles, while vice versa in blue bubbles. The size of each bubble corresponds to the $-\log_{10}(p \text{ value})$.

Smoking, a well-established risk factor for numerous systemic conditions, assumes the preeminent position in our findings, underscoring its detrimental impact on periodontal health. Observational studies have consistently reported a positive association between smoking and the incidence and severity of periodontitis [33]. Potential mechanisms through which smoking increases the risk of periodontitis can be attributed to several factors. Smoking is known to impair immune responses, leading to a compromised ability to fight against periodontal pathogens [34]. Additionally, tobacco smoke components can directly damage the oral tissues, including the periodontal ligament and alveolar bone [35]. Additionally, we observed a null association between type 2 diabetes, fasting glucose, blood sugar, and periodontitis risk, which contrasts with findings from a recent study that reported significant associations using a similar two-sample MR approach [36]. One potential explanation lies in the differences in the number of SNPs used as IVs for the exposures. Our study utilized a larger set of SNPs, which may have enhanced the comprehensiveness of the

genetic proxies but also introduced heterogeneity in the IVs. This methodological difference may partially account for the discrepancies with the recent study. In contrast to the null findings for type 2 diabetes, we identified a significant causal association between HbA1c levels and periodontitis, which aligns with the findings of the recent study [36]. This suggests that long-term glycemic control, as reflected by HbA1c, may be a more critical determinant of periodontitis risk than short-term glycemic markers or the presence of type 2 diabetes. These findings emphasize the importance of smoking cessation and glycaemic control in periodontal health management.

The present study failed to observe a significant association between the remaining potential factors and the risk of periodontitis. Notably, several expected associations, such as between sleep duration and periodontitis, did not reach statistical significance in our MR analysis despite prior observational evidence suggesting potential links. This unexpected lack of association

could be attributed to several methodological and biological factors. First, insufficient statistical power remains a significant concern in MR studies [37]. The genetic instruments for certain exposures, including sleep duration, may explain only a small proportion of phenotypic variance, limiting our ability to detect modest causal effects. Second, it is undeniable that certain characteristics can be altered, and they are strongly influenced by nongenetic factors that attenuate the impact on the risk of periodontitis, an aspect that may also appear to have no significant association with periodontitis [38, 39]. For instance, lifestyle choices such as alcohol consumption and dietary patterns are subject to considerable variability due to cultural, socioeconomic, and personal preferences. Another plausible explanation is that the null findings may indeed reflect a true absence of a causal relationship between these traits and periodontitis. Therefore, further research utilizing advanced methodologies, more comprehensive genetic instruments, and larger sample sizes is warranted to unravel the intricate interplay of these factors in the pathogenesis of periodontitis [40].

This is the first study, to the best of our knowledge, that conducted a comprehensive analysis using two-sample MR and meta-analyses to investigate the association of 50 potential risk factors and biomarkers with periodontitis. While our systematic evaluation of these known and suspected factors offers insights to aid clinicians, researchers, and policymakers in prioritizing risk factor interventions and developing personalized prevention and treatment plans, we also recognize that more experimental studies or alternative methods are needed to further strengthen this framework. Moreover, the identification of these factors opens new avenues for further MR and longitudinal studies, aimed at refining causal inference and elucidating underlying biological mechanisms to support personalized medicine approaches in periodontitis management.

However, there are some limitations that should be considered. First, the MR analysis relies on the assumption of no pleiotropy, which may not always hold true in complex biological systems. Importantly, our selection of genetic variants was primarily driven by statistical criteria rather than established biological pathways, which limits the mechanistic interpretation of SNP-exposure relationships. Second, we relaxed the significance thresholds used for IV selection in order to select more appropriately, potentially incorporating some of the instrumental variables with weaker associations with exposure factors. Moreover, due to the multiple exposures examined in our study, there exists a potential risk of type I error (false positives) arising from multiple testing, although we attempted to mitigate this through appropriate statistical corrections. Additionally, lifestyle and dietary factors might not be fully captured by the genetic proxies used in our analysis. These unmeasured confounders could potentially lead to an overestimation or underestimation of the true effect sizes. Third, to date, only a small fraction of genetic variants was available for several traits, such as birth weight and sex hormone-binding globulin. Therefore, risk estimates for these traits could be unstable or biased. Fourth, a common challenge in MR studies utilizing large GWAS datasets, including ours, is the reliance on the periodontitis case definitions available within those original datasets. These definitions may not perfectly align with current,

refined clinical classification schemes, representing a potential source of case misclassification bias that could affect the accuracy of our estimates. Moreover, the factors we studied may be primarily associated with periodontitis; this does not negate their relevance in all periodontal diseases, including those that progress rapidly. Fifth, although our study aims to highlight modifiable risk factors, the assumption of equivalence between genetic instruments and the environmental exposures they represent remains uncertain. Caution is warranted when extrapolating our findings to direct interventions; additional research is needed to determine whether genetically proxied exposures accurately capture modifiable aspects of these risk factors in practice. Furthermore, the exposure factors included in our study were multidimensional and numerous, and we conducted a preliminary exploration of potential risk factors for the occurrence and development of periodontitis. It is important to acknowledge that our nonsignificant results do not necessarily indicate the absence of causal relationships; rather, our study may have lacked sufficient statistical power to detect some genuine causal associations, particularly for exposures with subtle effects. We lacked strict and precise exposure selection criteria to conduct a multivariable MR study, which is an important direction for future research.

In conclusion, our study provides evidence for the potential causal association of various obesity-related indicators, smoking, HbA1c, and educational levels with the risk of periodontitis, highlighting the multifactorial nature of this complex disease. The identification of these factors may inform future preventive strategies and interventions aimed at reducing the burden of periodontitis in the population. Further research is warranted to elucidate the underlying mechanisms and causal relationships between these risk factors and periodontitis, as well as to explore the potential for personalized prevention and treatment based on individual risk profiles.

Author Contributions

Lingjun Yan contributed to the conception, project administration, methodology, validation, and writing the original draft. Zilin Liu contributed to the methodology, formal analysis, software, visualization, writing review, and editing. Bingqin Xie contributed to the methodology, validation, writing review, and editing. Yu Huang contributed to the methodology, data curation, writing review, and editing. Yuxuan Wu contributed to the formal analysis, software, writing review, and editing. Baochang He contributed to the methodology, visualization, writing review, and editing. Yanfen Li contributed to the conception, resources, writing review, and editing. Lan Luo contributed to the methodology, data curation, writing review, and editing. Fuhua Yan contributed to the conception, methodology, resources, writing review, and editing. Fa Chen contributed to the conception, methodology, writing review, and editing, supervision, project administration, and funding acquisition. Fa Chen gave the final approval of the version to be published. All authors have made substantial contributions to the conception and design of the study and gave their final approval and agree to be accountable for all aspects of the work.

Ethics Statement

The authors have nothing to report. Summary data were derived from previously reported studies from which ethical approval had been obtained.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in IEU open GWAS database at <https://gwas.mrcieu.ac.uk/>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.